

There are 3 types of math in resonance

- (i) Just applying formula
- (ii) slight modification of cut by source transformation $\rightarrow$ (Z) / (Y)
- (iii) No formula, Have to find Impedance / Admittance $\rightarrow$ (Z) / (Y)
 then set $\text{Im}(Z) = 0$ or $\text{Im}(Y) = 0$

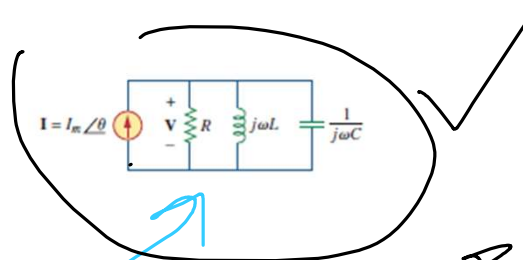
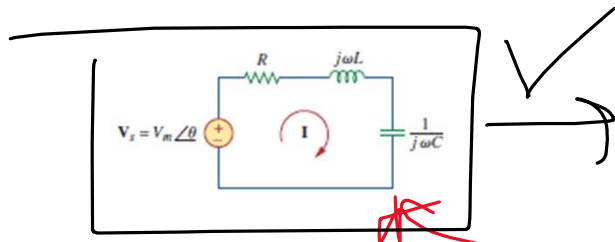
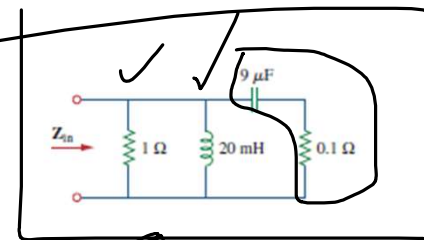
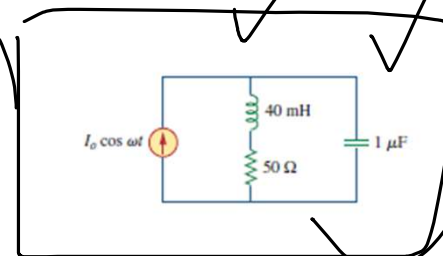


TABLE 14.4
Summary of the characteristics of resonant *RLC* circuits.

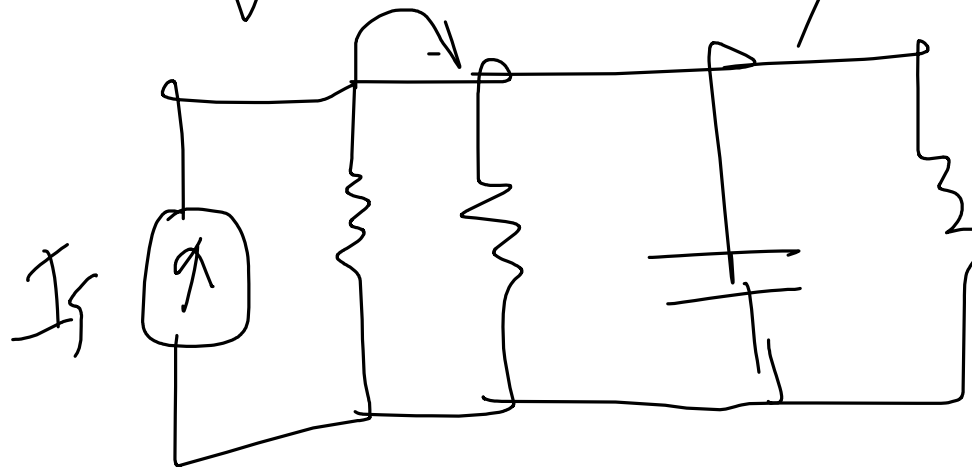
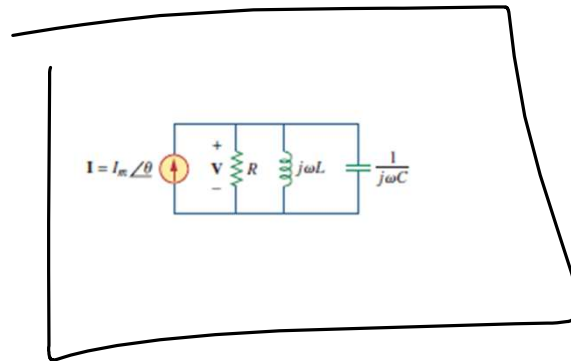
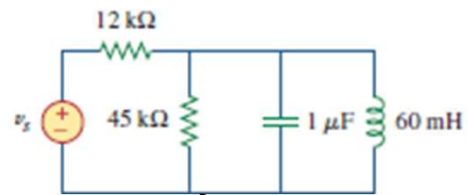
Characteristic	Series circuit	Parallel circuit
Resonant frequency, ω_0	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
Quality factor, Q	$\frac{\omega_0 L}{R}$ or $\frac{1}{\omega_0 RC}$	$\frac{R}{\omega_0 L}$ or $\omega_0 RC$
Bandwidth, B	$\frac{\omega_0}{Q}$	$\frac{\omega_0}{Q}$
Half-power frequencies, ω_1, ω_2	$\omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$	$\omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$
For $Q \geq 10$, ω_1, ω_2	$\omega_0 \pm \frac{B}{2}$	$\omega_0 \pm \frac{B}{2}$



Handwritten notes and calculations:

$I_m(\omega)$
 $= 0$

There are also handwritten circled numbers 4 and 7, and a checkmark.



type-2

14.29 Let $v_s = 20 \cos(\omega t)$ V in the circuit of Fig. 14.77. Find ω_0 , Q , and B , as seen by the capacitor.

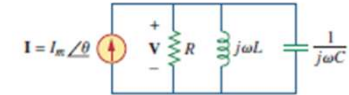
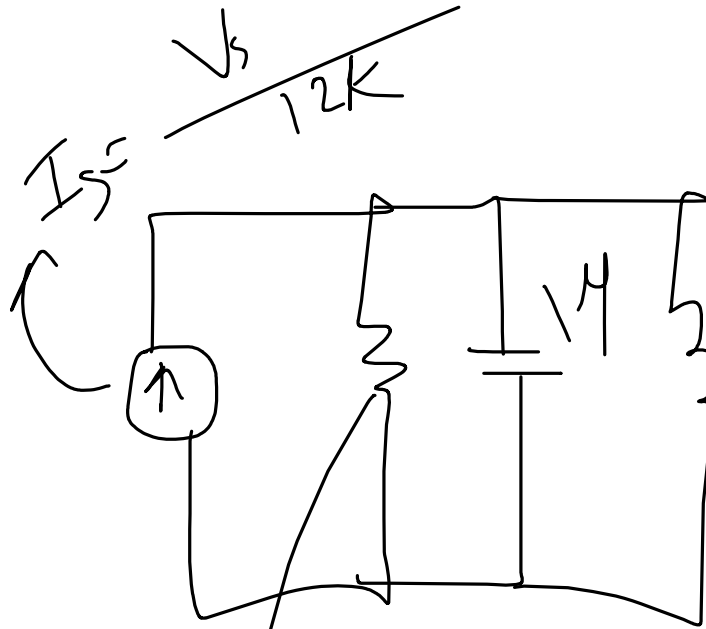
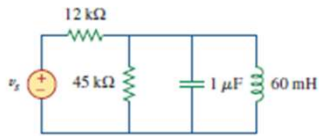


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$$\omega_0 =$$

$$\frac{1}{\sqrt{LC}}$$

$$= 4.082 \text{ k rad/s}$$

$$R \approx 12 \parallel 45$$

$$= 9.47 \text{ k}\Omega$$

60m

$$\boxed{\begin{aligned} \cancel{Q} &= \omega_0 R C \\ &= 38.66 \end{aligned}} \rightarrow \text{Unit-less}$$

$$B = \frac{\omega_0}{\cancel{Q}} =$$

type 2.

14.42 For the circuits in Fig. 14.81, find the resonant frequency ω_0 , the quality factor Q , and the bandwidth B .

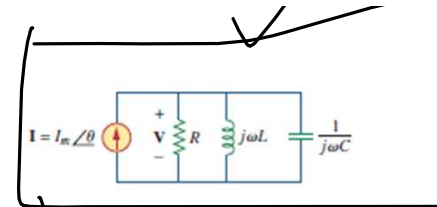
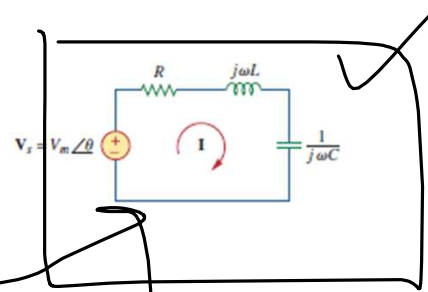
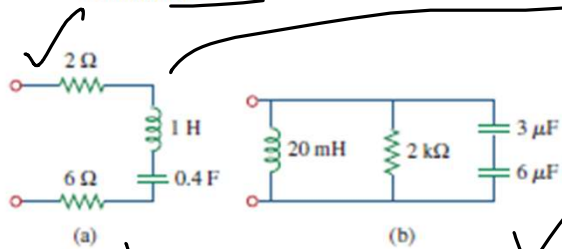
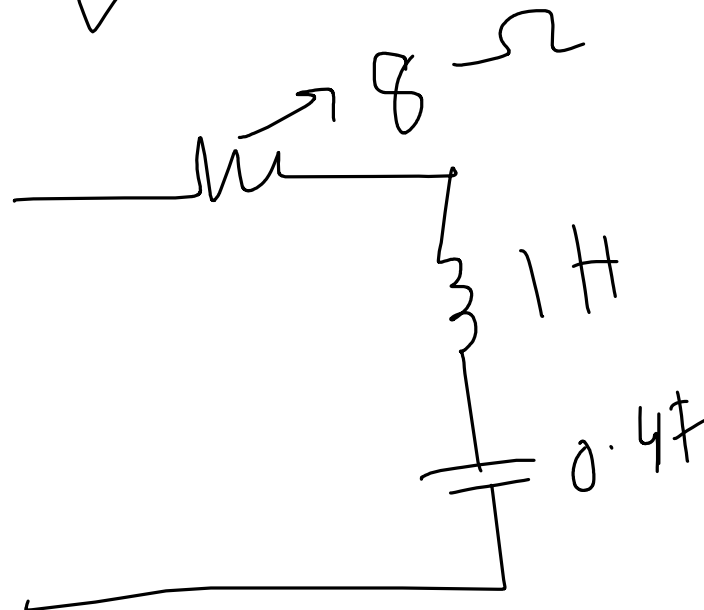


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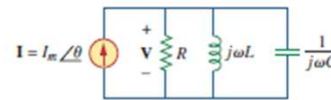


$$\omega_0 = \frac{1}{\sqrt{1 \times 0.4}}$$

$$= 1.58 \text{ rad/s}$$

$$Q = \frac{\omega_0 L}{R} = 0.198$$

$$\beta = \frac{\omega_0}{Q}$$



34, 64 series

$$\frac{3 \times 6}{3 + 6} = 2 \mu F$$

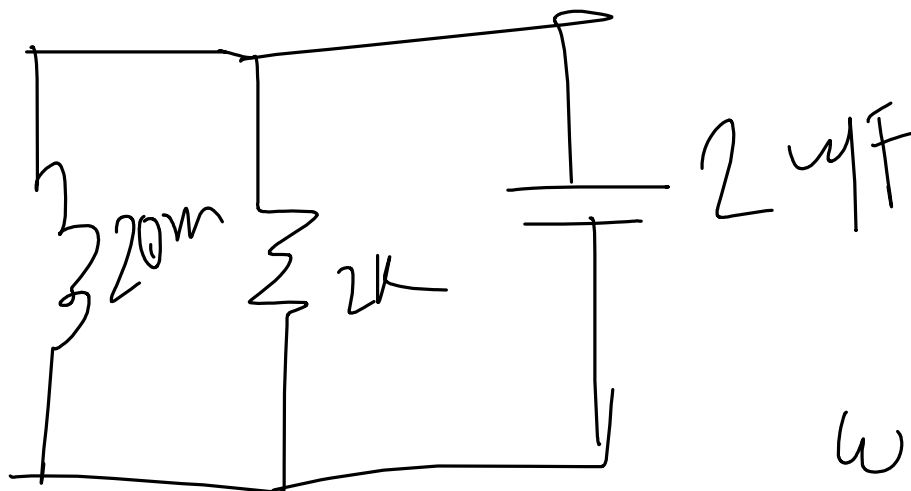


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$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$Q = \omega_0 RC$$

$$B = \frac{\omega_0}{Q}$$

type - 3

(1) Find Y

$$\text{Im}(Y) = 0$$

ω_0

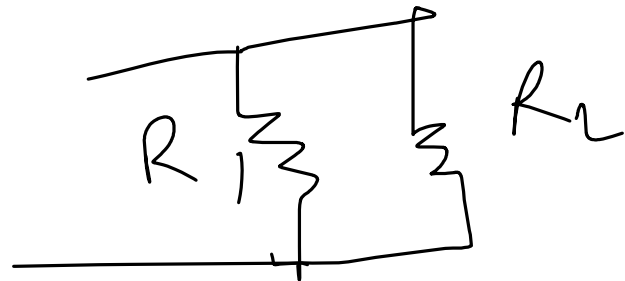
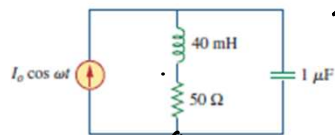
(2)

$$Y = \frac{1}{Z}$$

$$Z_0 = \left[\frac{1}{R_1} + \frac{1}{R_2} \right]$$

$$= \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$

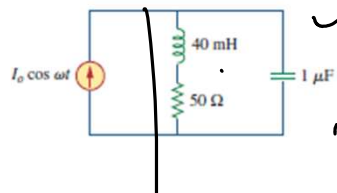
14.39 For the "tank" circuit in Fig. 14.79, find the resonant frequency.



$$Y_{10} = \frac{1}{R_1} + \frac{1}{R_2}$$

Type-3

14.39 For the "tank" circuit in Fig. 14.79, find the resonant frequency.



$$Z = (R + j\omega L) + \left(\frac{1}{j\omega C} \right)$$

$$Y = \frac{1}{R + j\omega L} + \left(\frac{1}{j\omega C} \right)$$

40 mH

~~40 mH~~

~~1 μF~~

~~(-)~~

$\sqrt{A + jB}$
 ω_0
 $\angle B = \delta$

Y

=

$$\frac{1}{R + j\omega L} + j\omega C$$

~~10~~ $(a + jb)(a - jb)$

$$Y = \frac{R - j\omega L}{(R + j\omega L)(R - j\omega L)} + j\omega C$$

$$= \frac{R - j\omega L}{R^2 - (j\omega L)^2} + j\omega C$$

$$= \frac{R - j\omega L}{R^2 + \omega^2 L^2} + j\omega C [j^2 = -1]$$

$$= \frac{R}{R^2 + \omega^2 L^2} - \frac{j\omega L}{R^2 + \omega^2 L^2} + j\omega L$$

$$Y = \frac{R}{R^2 + \omega^2 L^2} - j \left(\frac{\omega L}{R^2 + \omega^2 L^2} - \omega L \right)$$

$$A + jB$$

$$B = 0$$

At resonance

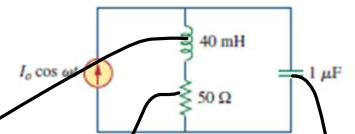
$$\text{Im}(Y) = 0$$

$$\Rightarrow \frac{\omega L}{R^2 + \omega^2 L^2} - \omega C = 0$$

$$\Rightarrow \omega L - \omega R^2 C - \omega^3 L^2 C = 0$$

$$\Rightarrow \omega(L - R^2 C - \omega^2 L^2 C) = 0$$

14.39 For the "tank" circuit in Fig. 14.79, find the resonant frequency.



$$L - R^2 C - \omega^2 L^2 C = 0$$

$L = 40 \text{ mH}$
 $R = 50 \Omega$
 $C = 1 \mu\text{F}$

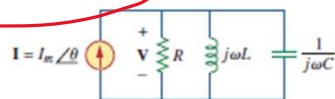
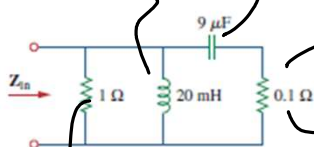
$$\omega^2 = \frac{L - R^2 C}{L^2 C} = 23.44 \times 10^6$$

$$\therefore \omega = 4.841 \text{ krad/s}$$

*14.44 For the circuit in Fig. 14.83, find:

(a) the resonant frequency ω_0

(b) $Z_{in}(\omega_0)$



type-3

$$\text{Im}(Z) = 0$$

$$Z_{in} = (R_1 \parallel j\omega L) \parallel \left(R_2 + \frac{1}{j\omega C} \right)$$

$$Z_{in} = \frac{R_1 j\omega L}{R_1 + j\omega L} \parallel \left(R_2 + \frac{1}{j\omega C} \right)$$

$$R_1 \parallel R_2 \rightarrow \frac{R_1 R_2}{R_1 + R_2}$$

$$\frac{R_a R_b}{R_a + R_b} =$$

$A + jB$
 $\downarrow$
 $B = 0$

$$\frac{R_1 j\omega L}{R_1 + j\omega L} \times \left(R_2 + \frac{1}{j\omega C} \right)$$

$$\frac{R_1 j\omega L}{R_1 + j\omega L} + R_2 + \frac{1}{j\omega C}$$

$$\frac{R_2 R_1 j\omega L}{R_1 + j\omega L} + \frac{R_1 j\omega L}{(R_1 + j\omega L)(j\omega C)}$$

$$\frac{(R_1 j\omega L)(j\omega C) + R_2 (R_1 + j\omega L)(j\omega C) + R_1 j\omega L}{(R_1 + j\omega L)(j\omega C)}$$

$$= \frac{R_1 R_2 j\omega L \times j\omega C + R_1 j\omega L}{\cancel{(R_1 + j\omega L) j\omega C} - R_1 \omega^2 LC + R_1 R_2 j\omega C + j\omega L R_2 C + R_1 j\omega L} \cancel{(R_1 + j\omega L) j\omega C}$$

(a-b)

$$= \frac{R_1 R_2 \omega^2 LC + R_1 j\omega L}{\underbrace{-R_1 \omega^2 LC + j(R_1 R_2 \omega C + \omega^2 L R_2 C + R_1 \omega L)}_b}$$

$$\begin{aligned}
 & \text{Im}(Z) = \frac{(\omega R_1 R_2 L C + j R_1 \omega L) \times (-R_1 \omega^2 L C - j R_1 \omega L + \omega^2 L R_2 C + R_1 \omega L)}{R_1^2 \omega^4 L^2 C^2 + (R_1 R_2 \omega C + \omega^2 L R_2 C + R_1 \omega L)^2} \\
 & \quad \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \\
 & \quad \quad \quad a^2 - b^2 \quad \quad \quad \text{cancel} \quad \quad \quad 2(amune)
 \end{aligned}$$

$$\text{Im}(Z) = 0$$

$$= \frac{-j R_1^2 \omega^3 L^2 C - j [R_1 R_2 \omega L + \omega^3 R_2 L C + R_1 \omega L] \times \omega R_2 R_2 C}{\omega R_2 R_2 C}$$

$$I_m(t) = \frac{-j (R_1^2 \omega^3 L^2 C + [\omega^3 R_1 R_2 L^2 + \omega^4 R_1 R_2^2 L^2 C^2 + \omega R_1 R_2^2 L^2])}{\omega R_2 R_2 C}$$

$$\operatorname{Im}(z) = 0$$

$$R \Rightarrow R_1^2 \omega^3 L^2 C + R_1^2 R_2^2 L C^2 + \omega^4 R_1 R_2 L^2 C^2 + \omega R_1 R_2 L^2 C = 0$$

$$\Rightarrow \omega^3 \times (R_1^2 L C + R_1^2 R_2 L C^2 + \omega R_1 R_2 L^2 C + R_1 R_2 L^2 C) = 0$$

$\hookrightarrow$ from this expression find ω